

Enumeration of linear codes with different hulls

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Abstract

The hull of a linear code is the intersection of the code with its dual and depends on the choice of inner product. In this paper, we study the enumeration of linear codes with prescribed Hermitian and symplectic hull dimensions. Extending earlier results on Euclidean hulls, we show that the numbers of codes with given Hermitian or symplectic hull dimensions satisfy inequalities analogous to those in the Euclidean case. Our approach relies on mass formulae for codes with fixed hull dimensions and provides a unified treatment of Euclidean, Hermitian, and symplectic hulls.

Key Words : Linear code, Hull, LCD code, Self-orthogonal code.

1 Introduction

Let C be a linear $[n, k]$ code over the finite field $\mathbb{F}_q$, that is, a k -dimensional linear subspace of the vector space $\mathbb{F}_q^n$. Assume that $\mathbb{F}_q^n$ is equipped with an inner product $\langle \cdot, \cdot \rangle$. The *dual code* (or *orthogonal code*) of C is defined by

$$C^\perp = \{u \in \mathbb{F}_q^n : \langle u, v \rangle = 0 \text{ for all } v \in C\}.$$

The intersection

$$\mathcal{H}(C) = C \cap C^\perp$$

is called the *hull* of C with respect to the given inner product. If $\mathcal{H}(C) = \{0\}$, then C is called an *LCD (Linear Complementary Dual) code*. If $C \subseteq C^\perp$, then C is said to be *self-orthogonal*.

In this work, we consider the following three inner products.

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- *Euclidean (standard) inner product:*

$$x \cdot y = \sum_{i=1}^n x_i y_i,$$

for any $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n) \in \mathbb{F}_q^n$.

- *Hermitian inner product:*

$$\langle x, y \rangle_H = \sum_{i=1}^n x_i y_i^q,$$

for any $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n) \in \mathbb{F}_{q^2}^n$. This product is defined only over fields with q^2 elements, where q is a prime power.

- *Symplectic inner product:*

$$\langle x, y \rangle_S = \sum_{i=1}^n (x_i y_{n+i} - x_{n+i} y_i),$$

for any $x = (x_1, \dots, x_{2n}), y = (y_1, \dots, y_{2n}) \in \mathbb{F}_q^{2n}$. The symplectic inner product is defined only in vector spaces of even dimension.

For a linear code C , we denote by $\mathcal{H}_E(C)$, $\mathcal{H}_H(C)$, and $\mathcal{H}_S(C)$ its Euclidean, Hermitian, and symplectic hulls, respectively. To illustrate the structure of these hulls for a single code, we present the following example.

Example 1. Let C be a linear $[6, 2]$ code over $\mathbb{F}_4 = \{0, 1, \omega, \omega^2\}$, where $1 + \omega + \omega^2 = 0$, generated by

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & \omega & \omega^2 & 1 \end{pmatrix}.$$

The code C is Euclidean self-orthogonal, and hence $\mathcal{H}_E(C) = C$. The two rows of G are Hermitian orthogonal to each other, but the second row is not Hermitian orthogonal to itself. Consequently, $\mathcal{H}_H(C) = \langle (101110) \rangle$. On the other hand, the two rows are not symplectic orthogonal, and therefore $\mathcal{H}_S(C) = \{0\}$.

Hence C is Euclidean self-orthogonal and symplectic LCD.

This work continues the authors' previous research on the enumeration of linear codes of fixed length and dimension over a finite field with prescribed Euclidean hull dimension. In [2], we studied the sequence

$$\mathcal{S}_{n,k,q} = (A_{n,k,0,q}, A_{n,k,1,q}, \dots, A_{n,k,\ell,q}, \dots, A_{n,k,k,q}),$$

where $A_{n,k,\ell,q}$ denotes the number of linear $[n, k]_q$ codes (equivalently, $[n, n-k]_q$ codes) whose Euclidean hull has dimension ℓ . Our main result in [2] is the following theorem.

Theorem 1. [2, Theorem 12] Let $\ell + 1 \leq k \leq n - \ell - 1$. Then $A_{n,k,\ell,q} > (q^{\ell+1} - 1)A_{n,k,\ell+1,q}$ for all possible values of q, n, k and ℓ except when q is odd, $n \equiv 2 \pmod{4}$, -1 is not a square in $\mathbb{F}_q^*$, and $k - \ell$ is odd, in which case $A_{n,k,\ell,q} \geq \frac{q^{\ell+1}-1}{2} A_{n,k,\ell+1,q}$.

We expect that an analogous dependence also holds for Hermitian and symplectic hulls. In particular, we show that the enumeration of linear codes with respect to the dimensions of their Hermitian and symplectic hulls exhibits behavior similar to that observed in the Euclidean case. By developing suitable adaptations of the counting techniques, we derive corresponding results in these two settings, thereby providing a unified treatment of Euclidean, Hermitian, and symplectic hulls.

In [4], the authors established mass formulae for linear codes with prescribed Hermitian and symplectic hull dimensions and obtained asymptotic results. Using these formulae, we prove results analogous to Theorem 1.

The rest of this extended abstract is organized as follows. Section 2 is devoted to the Hermitian hull of a linear code over a field with q^2 elements. It is proved in [3] that for $q > 2$ any linear code over $\mathbb{F}_{q^2}$ is monomially equivalent to an Hermitian LCD code. In Section 3, we consider symplectic inner product. As in the Euclidean and Hermitian cases, we prove that the number of linear codes with fixed length $2n$ and dimension $k \leq n$ decreases when the dimension of the symplectic hull increases.

2 Hermitian hulls

Let $\mathbb{F}_{q^2}$ be a finite field with q^2 elements and $\mathbb{F}_{q^2}^n$ be the n -dimensional vector space over this field. The following mass formula was proved in [4]. Here $HHull[n, k, \ell]_{q^2}$ denotes the number of all linear $[n, k]$ codes over $\mathbb{F}_{q^2}$ whose Hermitian hull has dimension ℓ .

Theorem 2. [4] Let n, k, ℓ be three positive integers such that $\ell \leq k \leq n - \ell$. Let $k_0 = k - \ell$. If $D_{n,k_0,i} = \prod_{i=1}^{k_0} \frac{q^{n-k_0+i} - (-1)^{n-k_0+i}}{q^i - (-1)^i}$, then

$$HHull[n, k, \ell]_{q^2} = \begin{cases} \left(\prod_{i=1}^{\ell} A_{n,k_0,i} \right) q^{k_0(n-k_0)} D_{n,k_0,i}, & \text{if } n - k_0 \text{ is odd,} \\ \left(\prod_{i=1}^{\ell} B_{n,k_0,i} \right) q^{k_0(n-k_0)} D_{n,k_0,i}, & \text{if } n - k_0 \text{ is even,} \end{cases}$$

where $A_{n,k_0,i} = \frac{(q^{m_i+2}+1)(q^{m_i+1}-1)}{q^{2k_0}(q^{2i}-1)}$ and $B_{n,k_0,i} = \frac{(q^{m_i+2}-1)(q^{m_i+1}+1)}{q^{2k_0}(q^{2i}-1)}$, $m_i = n - k_0 - 2i$.

We use a more convenient formula, given in the following corollary

Corollary 3. Let n, k, ℓ be positive integers such that $\ell \leq k \leq n - \ell$, and $n_0 = n - k$. Then

$$HHull[n, k, \ell]_{q^2} = \left(\prod_{i=1}^{\ell} M_{n, k, \ell, i} \right) q^{(k-\ell)(n_0+\ell)} \prod_{i=1}^{k-\ell} \frac{q^{n_0+\ell+i} - (-1)^{n_0+\ell+i}}{q^i - (-1)^i}, \quad (1)$$

where

$$M(n, k, \ell, i) = \frac{(q^{n_0+\ell-2i+2} - (-1)^{n_0+\ell})(q^{n_0+\ell-2i+1} + (-1)^{n_0+\ell})}{q^{2k-2\ell}(q^{2i} - 1)}.$$

Proof. Obviously, $n_0 + \ell = n - k + \ell = n - k_0$. Let

$$M(n, k, \ell, i) = \begin{cases} A_{n, k_0, i}, & \text{if } n - k_0 = n_0 + \ell \text{ is odd,} \\ B_{n, k_0, i}, & \text{if } n - k_0 = n_0 + \ell \text{ is even.} \end{cases}$$

Since

$$\begin{aligned} A_{n, k_0, i} &= \frac{(q^{n-k_0-2i+2} + 1)(q^{n-k_0-2i+1} - 1)}{q^{2k_0}(q^{2i} - 1)} \\ &= \frac{(q^{n_0+\ell-2i+2} + 1)(q^{n_0+\ell-2i+1} - 1)}{q^{2k-2\ell}(q^{2i} - 1)}, \end{aligned}$$

and

$$\begin{aligned} B_{n, k_0, i} &= \frac{(q^{n-k_0-2i+2} - 1)(q^{n-k_0-2i+1} + 1)}{q^{2k_0}(q^{2i} - 1)} \\ &= \frac{(q^{n_0+\ell-2i+2} - 1)(q^{n_0+\ell-2i+1} + 1)}{q^{2k-2\ell}(q^{2i} - 1)}, \end{aligned}$$

we have

$$M(n, k, \ell, i) = \frac{(q^{n_0+\ell-2i+2} - (-1)^{n_0+\ell})(q^{n_0+\ell-2i+1} + (-1)^{n_0+\ell})}{q^{2k-2\ell}(q^{2i} - 1)}.$$

Hence

$$\begin{aligned} HHull[n, k, \ell]_{q^2} &= \left(\prod_{i=1}^{\ell} M_{n, k, \ell, i} \right) q^{k_0(n-k_0)} \prod_{i=1}^{k_0} \frac{q^{n-k_0+i} - (-1)^{n-k_0+i}}{q^i - (-1)^i} \\ &= \left(\prod_{i=1}^{\ell} M_{n, k, \ell, i} \right) q^{(k-\ell)(n_0+\ell)} \prod_{i=1}^{k-\ell} \frac{q^{n_0+\ell+i} - (-1)^{n_0+\ell+i}}{q^i - (-1)^i}. \end{aligned}$$

□

When $\ell = 0$, the following formula is also proved in [4].

Theorem 4. [4] *The number of Hermitian LCD $[n, k]_{q^2}$ codes is*

$$HHull[n, k, 0]_{q^2} = q^{k(n-k)} \prod_{i=1}^k \frac{q^{n-k+i} - (-1)^{n-k+i}}{q^i - (-1)^i}. \quad (2)$$

In the rest of this section we build our investigation on Corollary 3 and Theorem 4. Compare first $HHull[n, k, 0]_{q^2}$ with $HHull[n, k, 1]_{q^2}$ for $1 \leq k \leq n - k$. Then

$$\frac{HHull[n, k, 0]_{q^2}}{HHull[n, k, 1]_{q^2}} = \frac{q^{n-1}(q^2 - 1)}{(q^{n_0} - (-1)^{n_0})(q^k - (-1)^k)}.$$

To evaluate this ratio, we use that $q^s - 1 < q^s$, $\frac{q^s+1}{q^s-1} < q + 1$. This gives us that

$$\frac{HHull[n, k, 0]_{q^2}}{HHull[n, k, 1]_{q^2}} = \begin{cases} \frac{\frac{q^{n-1}(q^2-1)}{(q^{n-k}-1)(q^k-1)}}{\frac{q^{n-1}(q^2-1)}{(q^{n-k}+1)(q^k+1)}} > \frac{q^2-1}{q} > q-1 & \text{if } n \text{ and } k \text{ are even} \\ \frac{q^{n-1}(q^2-1)}{(q^{n-k}+1)(q^k+1)} & \text{if } n \text{ is even and } k \text{ is odd} \\ \frac{\frac{q^{n-1}(q^2-1)}{(q^{n-k}+1)(q^k-1)}}{\frac{q^{n-1}(q^2-1)}{(q^{n-k}-1)(q^k+1)}} > \frac{q^2-1}{q+1} = q-1 & \text{if } n \text{ is odd and } k \text{ is even} \\ \frac{q^{n-1}(q^2-1)}{(q^{n-k}-1)(q^k+1)} > \frac{q^2-1}{q+1} = q-1 & \text{if } n \text{ and } k \text{ are odd.} \end{cases}$$

In the second case, if $k = 1$ then

$$\frac{HHull[n, k, 0]_{q^2}}{HHull[n, k, 1]_{q^2}} = \frac{q^{n-1}(q^2 - 1)}{(q^{n-1} + 1)(q + 1)} = \frac{q^{n-1}(q - 1)}{(q^{n-1} + 1)} < q - 1,$$

but is bigger than $q - 2$. Furthermore, if $k \geq 3$ then

$$\frac{HHull[n, k, 0]_{q^2}}{HHull[n, k, 1]_{q^2}} \geq \frac{q^5(q^2 - 1)}{(q^3 + 1)^2} = q - 1 + \frac{q^3(q^2 - 2) - 1}{(q^3 + 1)^2} > q - 1.$$

Hence, in all cases but $k = 1$ and even n it holds $\frac{HHull[n, k, 0]_{q^2}}{HHull[n, k, 1]_{q^2}} > q - 1$.

The calculations in the case $0 < \ell < k$ give us that

$$\frac{HHull[n, k, \ell]_{q^2}}{HHull[n, k, \ell + 1]_{q^2}} = \frac{q^{n-2\ell-1}(q^{2\ell+2} - 1)}{(q^{n-k-\ell} - (-1)^{n-k+\ell})(q^{k-\ell} - (-1)^{k-\ell})}.$$

There are 8 different possibilities for the triple $(n \bmod 2, k \bmod 2, \ell \bmod 2)$, which we group in 4 cases. Recall that we consider only the codes with $n \geq 2k$, and if n is odd, then $n \geq 2k + 1$.

- Let n and $k - \ell$ be even. Then

$$\frac{HHull[n, k, \ell]_{q^2}}{HHull[n, k, \ell + 1]_{q^2}} = \frac{q^{n-2\ell-1}(q^{2\ell+2} - 1)}{(q^{n-k-\ell} - 1)(q^{k-\ell} - 1)} > \frac{q^{2\ell+2} - 1}{q} > q^{2\ell+1} - 1.$$

- If n is odd, but $k - \ell$ is even, then $n_0 + \ell = n - (k - \ell)$ is odd and

$$\frac{HHull[n, k, \ell]_{q^2}}{HHull[n, k, \ell + 1]_{q^2}} = \frac{q^{n-2\ell-1}(q^{2\ell+2} - 1)}{(q^{n-k-\ell} + 1)(q^{k-\ell} - 1)}.$$

Let's take a closer look at the denominator:

$$(q^{n-k-\ell} + 1)(q^{k-\ell} - 1) = q^{n-2\ell} + q^{k-\ell}(1 - q^{n-2k}) - 1 < q^{n-2\ell}.$$

It follows that

$$\frac{HHull[n, k, \ell]_{q^2}}{HHull[n, k, \ell + 1]_{q^2}} > \frac{q^{2\ell+2} - 1}{q} > q^{2\ell+1} - 1.$$

- Let n and $k - \ell$ be odd. In this case $n_0 + \ell$ is even and so

$$\frac{HHull[n, k, \ell]_{q^2}}{HHull[n, k, \ell + 1]_{q^2}} = \frac{q^{n-2\ell-1}(q^{2\ell+2} - 1)}{(q^{n-k-\ell} - 1)(q^{k-\ell} + 1)} > \frac{q^{2\ell+2} - 1}{q + 1} > q^{2\ell+1} - q^{2\ell}.$$

- If n is even, but $k - \ell$ is odd, then

$$\frac{HHull[n, k, \ell]_{q^2}}{HHull[n, k, \ell + 1]_{q^2}} = \frac{q^{n-2\ell-1}(q^{2\ell+2} - 1)}{(q^{n-k-\ell} + 1)(q^{k-\ell} + 1)} \geq \frac{q(q^{2\ell+2} - 1)}{(q + 1)^2} > q^{2\ell+1} - q^{2\ell}.$$

The following theorem is the main result of this section, which is proved by the arguments above.

Theorem 5. *Let $1 \leq \ell + 1 \leq k \leq n - k$. Then*

$$HHull[n, k, \ell]_{q^2} > \begin{cases} (q^{\ell+1} - 1)HHull[n, k, \ell + 1]_{q^2} & \text{if } k - \ell \text{ is odd,} \\ q^{2\ell}(q - 1)HHull[n, k, \ell + 1]_{q^2} & \text{if } k - \ell \text{ is even.} \end{cases}$$

3 Symplectic hulls

The symplectic inner product is defined by

$$\langle x, y \rangle_S = \sum_{i=1}^n (x_i y_{n+i} - x_{n+i} y_i) = x \Omega_n y^T,$$

where $x = (x_1, \dots, x_{2n})$, $y = (y_1, \dots, y_{2n}) \in \mathbb{F}_q^{2n}$, and $\Omega_n = \begin{pmatrix} O & I_n \\ -I_n & O \end{pmatrix}$.

Proposition 6. *[7, Proposition 2] Let C be a linear $[2n, k]_q$ code with a generator matrix G . Then C has ℓ -dimensional symplectic hull if and only if $\ell = k - \text{rank}(G \Omega_n G^T)$.*

We need also the following proposition.

Proposition 7. [4, Proposition 4.4] *If there is a linear $[2n, k]_q$ code with ℓ -dimensional symplectic hull, then $k - \ell$ is even.*

According to this proposition, symplectic LCD codes exist only if k is even. We denote the number of all linear $[2n, k]_q$ codes, whose symplectic hull has dimension ℓ , by $SHull[2n, k, \ell]_q$.

Theorem 8. [4, Theorem 4.9] *Assume that n, k and ℓ are three nonnegative integers with $\ell \leq k \leq 2n$ and $k = 2k_0 + \ell$. Then*

$$SHull[2n, k, \ell]_q = \left(\prod_{i=1}^{\ell} \frac{q^{2n-2k_0-2i+2} - 1}{q^{2k_0+i} - q^{2k_0}} \right) q^{2k_0(n-k_0)} \begin{bmatrix} n \\ k_0 \end{bmatrix}_{q^2}. \quad (3)$$

In the above theorem, $\begin{bmatrix} n \\ k_0 \end{bmatrix}_{q^2}$ is the Gaussian binomial coefficient, defined by $\begin{bmatrix} n \\ 0 \end{bmatrix}_{q^2} = 1$,

$$\begin{bmatrix} n \\ k_0 \end{bmatrix}_{q^2} = \frac{(q^{2n} - 1)(q^{2(n-1)} - 1) \dots (q^{2(n-k_0+1)} - 1)}{(q^{2k_0} - 1)(q^{2(k_0-1)} - 1) \dots (q^2 - 1)}, \text{ if } 1 \leq k_0 \leq n.$$

Again, as in Section 2, we use a more convenient formula presented in the following corollary.

Corollary 9. *Assume that n, k and ℓ are three nonnegative integers with $\ell \leq k \leq n$ and $k = 2k_0 + \ell$. If $\ell \geq 1$ then*

$$SHull[2n, k, \ell]_q = q^{2k_0(n-k_0-\ell)} \begin{bmatrix} n \\ k_0 \end{bmatrix}_{q^2} \prod_{i=1}^{\ell} \frac{q^{2n-2k_0-2i+2} - 1}{q^i - 1}. \quad (4)$$

Furthermore, if k is even, then

$$SHull[2n, k, 0]_q = q^{k(n-k/2)} \begin{bmatrix} n \\ k/2 \end{bmatrix}_{q^2}. \quad (5)$$

In the rest of this section, as in Section 2, we build our investigation on Corollary 9. Note that Corollary 9 also covers the case $\ell = 0$, which is different from Corollary 3 of Section 2.

Let $\ell \leq k - 2$ and $\ell_1 = \ell + 2$. If $k_1 = k_0 - 1$ then $k = 2k_1 + \ell_1$. As in the previous section, we evaluate the ratio $SHull[2n, k, \ell]_q / SHull[2n, k, \ell_1]_q$. For that, we need

$$\begin{aligned} \frac{\begin{bmatrix} n \\ k_0 \end{bmatrix}_{q^2}}{\begin{bmatrix} n \\ k_1 \end{bmatrix}_{q^2}} &= \frac{\begin{bmatrix} n \\ k_0 \end{bmatrix}_{q^2}}{\begin{bmatrix} n \\ k_0 - 1 \end{bmatrix}_{q^2}} = \frac{q^{2(n-k_0+1)} - 1}{q^{2k_0} - 1} = \frac{q^{2n-k+\ell+2} - 1}{q^{k-\ell} - 1} \\ \frac{q^{2k_0(n-k_0-\ell)}}{q^{2k_1(n-k_1-\ell_1)}} &= q^{2k_0(n-k_0-\ell)-2k_1(n-k_1-\ell_1)} = q^{2n-2\ell-2} \end{aligned}$$

Let us first compare $SHull[2n, k, 0]_q$ with $SHull[2n, k, 2]_q$ for even dimension. In this case

$$\begin{aligned} \frac{SHull[2n, k, 0]_q}{SHull[2n, k, 2]_q} &= \frac{q^{2k_0(n-k_0)} \begin{bmatrix} n \\ k_0 \end{bmatrix}_{q^2} (q-1)(q^2-1)}{q^{2k_1(n-k_1-2)} \begin{bmatrix} n \\ k_1 \end{bmatrix}_{q^2} (q^{2n-2k_1}-1)(q^{2n-2k_1-2}-1)} \\ &= \frac{q^{2n-2} (q^{2n-k+2}-1)(q-1)(q^2-1)}{(q^k-1)(q^{2n-k+2}-1)(q^{2n-k}-1)} \\ &= \frac{q^{2n-2} (q-1)(q^2-1)}{(q^k-1)(q^{2n-k}-1)} > \frac{(q-1)(q^2-1)}{q^2}. \end{aligned}$$

For $\ell \geq 1$ it holds

$$\begin{aligned} \frac{SHull[2n, k, \ell]_q}{SHull[2n, k, \ell_1]_q} &= \frac{q^{2n-2\ell-2} (q^{2n-k+\ell+2}-1)(q^{\ell+1}-1)(q^{\ell+2}-1) \prod_{i=1}^{\ell} (q^{2n-2k_0-2i+2}-1)}{(q^{k-\ell}-1) \prod_{i=1}^{\ell+2} (q^{2n-2k_1-2i+2}-1)} \\ &= \frac{q^{2n-2\ell-2} (q^{2n-k+\ell+2}-1)(q^{\ell+1}-1)(q^{\ell+2}-1)}{(q^{k-\ell}-1)(q^{2n-2k_0+2}-1)(q^{2n-2k_0-2\ell}-1)} \\ &= \frac{q^{2n-2\ell-2} (q^{2n-k+\ell+2}-1)(q^{\ell+1}-1)(q^{\ell+2}-1)}{(q^{k-\ell}-1)(q^{2n-k+\ell+2}-1)(q^{2n-k-\ell}-1)} \\ &= \frac{q^{2n-2\ell-2} (q^{\ell+1}-1)(q^{\ell+2}-1)}{(q^{k-\ell}-1)(q^{2n-k-\ell}-1)} \\ &> \frac{(q^{\ell+1}-1)(q^{\ell+2}-1)}{q^2}. \end{aligned}$$

This proves the following theorem

Theorem 10. Assume that n , k and ℓ are three nonnegative integers with $0 \leq \ell \leq k-2$, $1 \leq k \leq n$, and $k-\ell$ is even. Then

$$SHull[2n, k, \ell]_q > \frac{(q^{\ell+1}-1)(q^{\ell+2}-1)}{q^2} SHull[2n, k, \ell+2]_q.$$

Example 2. Let $q = 3$, $n = 5$ and $k = 3$. Then

$$SHULL[10, 3, 1]_3 = 17,648,856,720, \quad SHULL[10, 3, 3]_3 = 677,871,040.$$

It is good to mention that the number of Euclidean self-orthogonal $[10, 3, 3]_3$ codes is 22,408,960.

Remark 1. An interesting fact is that the number of symplectic self-orthogonal $[2n, k]_q$ codes is $\pi_{2n,k,q}$, where the function $\pi_{2n,k,q} = \prod_{i=1}^k \frac{q^{2n-2i+2}-1}{q^i-1}$ is used in the formula for the number of Euclidean self-orthogonal codes $\sigma_{2n,k,q}$ proved by Vera Pless in [5], namely

$$\sigma_{2n,k,q} = \begin{cases} \frac{q^{2n-k}-1}{q^{2n}-1} \pi_{2n,k,q}, & \text{if } q \text{ is even, } k \leq n \\ \frac{q^{n-k}+1}{q^n+1} \pi_{2n,k,q}, & \text{if } q \text{ is odd and } (-1)^n \text{ is a square, } k \leq n \\ \frac{q^{n-k}-1}{q^n-1} \pi_{2n,k,q}, & \text{if } q \text{ is odd and } (-1)^n \text{ is not a square, } k \leq n-1. \end{cases}$$

These formulas show that the symplectic self-orthogonal $[2n, k]_q$ codes are much more than the Euclidean self-orthogonal codes with the same parameters. For the self-dual codes in even q we have $\pi_{2n, n, q} = (q^n + 1)\sigma_{2n, n}$.

4 Conclusion

In this extended abstract, we investigated the enumeration of linear codes with prescribed Hermitian and symplectic hull dimensions, extending known results for Euclidean hulls. Several directions for future research emerge from our results. One such direction is a systematic comparison of the numbers of linear codes with the same length and dimension but different types of hulls, such as Euclidean, Hermitian, and symplectic hulls. Understanding how the choice of inner product affects the distribution of hull dimensions for fixed code parameters may reveal deeper structural differences between these classes of codes.

Another promising direction is to complement the theoretical results of this paper with a computational classification of quaternary linear codes with respect to the Hermitian inner product. In particular, we may study the distribution of Hermitian hull dimensions among inequivalent quaternary codes of fixed length and dimension, with special emphasis on Hermitian LCD and Hermitian self-orthogonal codes. For this purpose, we will use the program GENERATION from the software package QEXTNEWEDITION [1]. We focus on the quaternary case since $\mathbb{F}_4$ is the only finite field for which the equivalence relation preserves the Hermitian inner product which means that equivalent codes have equivalent Hermitian hulls.

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References

- [1] I. Bouyukliev, The program Generation in the software package QextNewEdition. In: A. Bigatti, J. Carette, J. Davenport, M. Joswig, T. de Wolff (eds), *Mathematical Software – ICMS 2020, Lecture Notes in Computer Science*, vol. 12097, Springer, Cham (2020).
- [2] S. Bouyuklieva, I. Bouyukliev, F. Özbudak, Sequence of Numbers of Linear Codes with Increasing Hull Dimensions, to appear in *Designs, Codes and Cryptography*, (2026).

- [3] C. Carlet, S. Mesnager, C. Tang, Y. Qi and R. Pellikaan, Linear codes over $\mathbb{F}_q$ are equivalent to LCD codes for $q > 3$, *IEEE Trans. Inform. Theory*, **64**, (2018) 3010–3017.
- [4] Shitao Li, Minjia Shi, Yang Li, San Ling, A further study on the mass formula for linear codes with prescribed hull dimension, arXiv preprint arXiv:2410.13578, (2024).
- [5] Vera Pless, On the uniqueness of the Golay codes, *Journal of Combinatorial Theory*, **5(3)**, (1968) 215–228.
- [6] Nicolas Sendrier, On the Dimension of the Hull, *SIAM J. Discret. Math.* **10(2)**, (1997) 282–293.
- [7] H. Xu, W. Du, On some binary symplectic self-orthogonal codes, *Applicable Algebra in Engineering, Communication and Computing*, **33**, (2022) 321–337.